CLASS 8 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A block is pulled with 4 N but does not move. The friction acting on it is: 1
(a) 2 N **(b)** 4 N **(c)** 8 N **(d)** 0 N

2. A force of 240 N acts on an area of 0.4 square metre. The pressure is: 1
(a) 240 Pa **(b)** 60 Pa **(c)** 96 Pa **(d)** 600 Pa

3. A crystal of potassium permanganate is dropped into water in two beakers, one hot and one cold. The colour spreads faster in the hot beaker because: 1
(a) crystal dissolves into gas
(b) particles move faster at higher temperature
(c) water gets heavier
(d) cold water is thicker

4. Which of the following pairs contains only elements? 1
(a) CO_2 and N_2 **(b)** H_2O and O_2
(c) $NaCl$ and Fe **(d)** O_2 and Fe

5. 40 g of salt is added to 160 g of water. The mass percentage of the solution is: 1
(a) 10 per cent **(b)** 20 per cent **(c)** 40 per cent **(d)** 25 per cent

6. A glass rod is rubbed with silk and brought near a suspended charged plastic rod, which is repelled. The plastic rod has: 1
(a) magnetism **(b)** negative charge **(c)** no charge **(d)** positive charge

7. Air pressure inside a bicycle tyre is greater than outside because: 1
(a) tyre is hot **(b)** rubber is thick
(c) more air is pumped in **(d)** the valve is open

8. Steam at 100 degrees Celsius has more energy than water at 100 degrees Celsius because it has: 1
(a) latent heat of vaporisation **(b)** more particles
(c) more mass **(d)** lower temperature

9. Which of the following is the odd one out? 1
(a) sulphur **(b)** iron **(c)** mercury **(d)** copper

10. A beam of light passes through a mixture and becomes visible. The mixture is a: 1

(a) true solution **(b)** pure compound **(c)** colloid **(d)** gas

11. Which of the following is NOT an effect of force? 1

(a) change in colour **(b)** change in speed
(c) change in shape **(d)** change in direction

12. A hurricane in the Atlantic and a typhoon in the Pacific are: 1

(a) tornadoes
(b) the same type of storm with different names
(c) earthquakes
(d) different types of storms

13. A gas at high pressure and low temperature is more likely to become: 1

(a) plasma **(b)** liquid **(c)** solid always **(d)** vapour only

14. A bell is made of bronze because it is: 1

(a) liquid **(b)** sonorous and hard
(c) brittle **(d)** soft

15. A student prepares tea with 5 g sugar in 100 mL and another with 10 g sugar in 100 mL. The second tea is: 1

(a) saturated **(b)** more dilute
(c) more concentrated **(d)** a suspension

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A trolley is pushed with 15 N and the friction is 6 N. Find the net force and say in which direction it acts. 2

17. A box exerts a force of 80 N on a floor over 0.2 square metre. Find the pressure. 2

18. Why does the volume of a mixture of alcohol and water appear less than the sum of their volumes? 2

19. Why is water called a compound and not a mixture? Give two reasons. 2

20. Find the mass percentage of a solution made by dissolving 18 g of sugar in 82 g of water. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A box of mass 10 kg is on a floor. A spring balance shows 25 N just before it starts moving and 18 N while sliding. Explain the readings. 3

22. A girl of weight 400 N stands on one foot of area 0.02 square metre. Find the pressure and compare with standing on two feet. 3

23. Explain why solids retain their shape, liquids flow and gases spread using particle ideas. 3

24. A student mixed iron filings and sulphur and heated them. Write three observations and the inference. 3

OR

A solution is prepared by dissolving 35 g of salt in 140 g of water. Find the mass of the solution and the mass percentage of salt. 3

25. Explain why the weight of an astronaut changes on the Moon but not the mass. 3

OR

Differentiate between a cyclone, a tornado and a thunderstorm (one point each). 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A student pulls a block with a spring balance on three surfaces: polished table (2 N), wooden floor (5 N) and carpet (9 N). Explain the observations and draw a conclusion. How will the readings change if a 1 kg mass is added? 5

OR

A tractor with weight 36000 N has four wheels each in contact with 0.3 square metre of ground. Find the pressure on the ground and compare with a person of weight 600 N standing on feet of area 0.05 square metre in total. 5

27. Describe an experiment to show that particles of matter are continuously moving and have spaces between them, using KMnO4 and water. 5

OR

Describe the properties of the iron-sulphur mixture and iron sulphide, and conclude what is seen about mixtures and compounds. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND